WorshipCue

Music stays personal. Preparation brings the team together.

DEVELOPMENT ARCHITECTURE

Stakeholder view | October 2026

01 MUSICIANS + GUESTS

Native iPad

SwiftUI / PDFKit / PencilKit
Charts, ink and local setlists
SQLite / GRDB persistence

Prepared music works offline

Browser guest

Self-hosted PDF.js viewer
Published music + group chat
Name entry; no installation

One room + one service
Expiring, revocable guest access

Neon Auth

Member email-code sign-in
Managed session verification

AUTH ONLY; AWS DATA STAYS

verify session

Sign-in is separate from team access.
Verified identity + an explicit invitation
are required for the selected team's content.

02 AWS TEAM SERVICES

Managed services; private data and files

Amazon API Gateway

HTTPS team requests
Foreground WebSocket hints
Both clients fetch updates

AWS Lambda

Python request handlers
Identity + team authorization
Validate before publication

Amazon DynamoDB

Teams, sets and messages
Note metadata + revisions
Durable state; on demand

Amazon S3

Private immutable PDFs
Drawing files + previews
Short-lived signed file URLs

Authorized file operations

LIVE UPDATE RULE

Hints trigger a state refresh.
Chat and cues never switch
the musician's open chart.

ACCESS CHECKS

Exact church + team membership
Personal ink: owner access only
Guest scope + revocation checks

Amazon Cognito

Guest identities; original
member sign-in route retained

Amazon SES

Original OTP email path
Restricted at last recorded check

03 RECOVERY + OPERATIONS

Backups and monitoring protect the team workspace.

Amazon EventBridge

Scheduled checkpoints resume the backup job.
At most one daily backup job is started.

Backup worker / AWS Lambda

DynamoDB backup + version-pinned S3 file copies
Verified manifest; restore into an isolated scratch table

Monitor + notify

CloudWatch alarms + SNS operator alerts
Backup failures/staleness and mail feedback

OFFLINE FIRST Local charts and personal notes

MUSICIAN CONTROL Manual song opening and page turns

TEAM ISOLATION Separate libraries and chat

Current development implementation. Capacity, hardware and release qualification remain open.

Official AWS / Neon / Swift assets | Aiden Rhaa